

# [Soal] Geometry Everyday Project

AZZAM LABIB HAKIM |INSTAGRAM: @HAXUV\_WORLD |THREADS: @AZZAM\_29\_12

September 8, 2026

## Daftar Isi

|                                                                         |          |
|-------------------------------------------------------------------------|----------|
| <b>1 Soal</b>                                                           | <b>3</b> |
| 1.1 Korea MO 2022, Problem 1 . . . . .                                  | 3        |
| 1.2 Japan MO 2022, Problem 3 . . . . .                                  | 3        |
| 1.3 IMO 2014, Problem 4 . . . . .                                       | 3        |
| 1.4 USA January TST for IMO 2016, Problem 3 . . . . .                   | 3        |
| 1.5 Singapore MO Open 2022 . . . . .                                    | 3        |
| 1.6 Iran-Ukraine Camp 2021 Exam I P-1 . . . . .                         | 4        |
| 1.7 USAMO 2015, Problem 2 . . . . .                                     | 4        |
| 1.8 All-Russian 2022 Grade 11 Problem 3 . . . . .                       | 4        |
| 1.9 All-Russian 2022 Grade 9 Problem 2 . . . . .                        | 4        |
| 1.10 CGMO 2022, Problem 3 . . . . .                                     | 5        |
| 1.11 IMSO 2021 . . . . .                                                | 5        |
| 1.12 Philippine MO 2022, Problem 6 . . . . .                            | 5        |
| 1.13 Finnish National High School Competition 2019, Problem 3 . . . . . | 5        |
| 1.14 Finnish National High School Competition 2001, Problem 1 . . . . . | 5        |
| 1.15 Iberoamerican 2022, Problem 5 . . . . .                            | 6        |
| 1.16 Random Problem by Wildan Bagus Wicaksono . . . . .                 | 6        |
| 1.17 USAJMO 2023, Problem 2 . . . . .                                   | 6        |
| 1.18 Korea 2023, Problem 1 . . . . .                                    | 6        |
| 1.19 IMO 2023, Problem 2 . . . . .                                      | 6        |
| 1.20 Japan 2023, Problem 2 . . . . .                                    | 7        |
| 1.21 Random Problem 2 . . . . .                                         | 7        |

|                                                                             |    |
|-----------------------------------------------------------------------------|----|
| 1.22 LMNAS SMP 33 Penyisihan nomor 16 . . . . .                             | 7  |
| 1.23 LMNAS SMP 33 Penyisihan nomor 3 isian singkat . . . . .                | 7  |
| 1.24 LMNAS SMP 31 Semifinal nomor 17 . . . . .                              | 7  |
| 1.25 Pelatda Pra OSP SMA 2024 DKI Jakarta . . . . .                         | 7  |
| 1.26 IMO 2024, Problem 4 . . . . .                                          | 8  |
| 1.27 Random Problem 3 . . . . .                                             | 8  |
| 1.28 KTOM Januari 2025, Nomor 3 Esai . . . . .                              | 8  |
| 1.29 OSN SMP 2016, Nomor 3, Hari Pertama . . . . .                          | 8  |
| 1.30 Random Problem 4 . . . . .                                             | 8  |
| 1.31 A lemma from Berkeley Math Tournament 2024 . . . . .                   | 9  |
| 1.32 Japan MO 2025, Problem 2 . . . . .                                     | 9  |
| 1.33 OSP SMA 2019, Essay, Problem 5 . . . . .                               | 9  |
| 1.34 British MO 2019 Round 2, Problem 1 . . . . .                           | 9  |
| 1.35 JBMO 2018, Problem 4 . . . . .                                         | 9  |
| 1.36 Singapore MO Open 2025, Problem 1 . . . . .                            | 10 |
| 1.37 Simulasi Semifinal OSN SMP 2025 DKI Jakarta, Uraian, Nomor 1 . . . . . | 10 |
| 1.38 Kompetisi Sains Ruangguru 2021 Nomor 1 . . . . .                       | 10 |
| 1.39 Iran MO 2018 Round 1, Problem 20 . . . . .                             | 10 |
| 1.40 Random Problem 5, Classical Incenter Altitude . . . . .                | 10 |
| 1.41 OSS (Seleksi Sekolah) Pradipta Dirgantara 2026 . . . . .               | 11 |

## §1 Soal

### §1.1 Korea MO 2022, Problem 1

Let  $ABC$  be an acute triangle with circumcenter  $O$ , and let  $D$ ,  $E$ , and  $F$  be the feet of altitudes from  $A$ ,  $B$ , and  $C$  to sides  $BC$ ,  $CA$ , and  $AB$ , respectively. Denote by  $P$  the intersection of the tangents to the circumcircle of  $ABC$  at  $B$  and  $C$ . The line through  $P$  perpendicular to  $EF$  meets  $AD$  at  $Q$ , and let  $R$  be the foot of the perpendicular from  $A$  to  $EF$ . Prove that  $DR$  and  $OQ$  are parallel.

### §1.2 Japan MO 2022, Problem 3

In an isosceles triangle  $ABC$ ,  $AB = AC$ . Take a point  $O$  inside the triangle (not on the boundary). Draw a circle  $\omega$  with center  $O$  and passing through  $C$ . Suppose  $\omega \cap BC = \{C, D\}$ ,  $\omega \cap AC = \{C, E\}$ . Denote the circumcircle of  $\triangle AEO$  as  $\Gamma$ . And suppose  $\Gamma \cap \omega = \{E, F\}$ . Prove that the circumcenter of  $\triangle BDF$  is on  $\Gamma$ .

### §1.3 IMO 2014, Problem 4

Let  $P$  and  $Q$  be on segment  $BC$  of an acute triangle  $ABC$  such that  $\angle PAB = \angle BCA$  and  $\angle CAQ = \angle ABC$ . Let  $M$  and  $N$  be the points on  $AP$  and  $AQ$ , respectively, such that  $P$  is the midpoint of  $AM$  and  $Q$  is the midpoint of  $AN$ . Prove that the intersection of  $BM$  and  $CN$  is on the circumference of triangle  $ABC$ .

### §1.4 USA January TST for IMO 2016, Problem 3

Let  $ABC$  be an acute scalene triangle and let  $P$  be a point in its interior. Let  $A_1$ ,  $B_1$ ,  $C_1$  be projections of  $P$  onto triangle sides  $BC$ ,  $CA$ ,  $AB$ , respectively. Find the locus of points  $P$  such that  $AA_1$ ,  $BB_1$ ,  $CC_1$  are concurrent and  $\angle PAB + \angle PBC + \angle PCA = 90^\circ$ .

### §1.5 Singapore MO Open 2022

For  $\triangle ABC$  and its circumcircle  $\omega$ , draw the tangents at  $B, C$  to  $\omega$  meeting at  $D$ . Let the line  $AD$  meet the circle with center  $D$  and radius  $DB$  at  $E$  inside  $\triangle ABC$ . Let  $F$  be the point on the extension of  $EB$  and  $G$  be the point on the segment  $EC$  such that  $\angle AFB = \angle AGE = \angle A$ . Prove that the tangent at  $A$  to the circumcircle of  $\triangle AFG$  is parallel to  $BC$ .

**§1.6 Iran-Ukraine Camp 2021 Exam I P-1**

Incircle of the triangle  $\triangle ABC$  is tangent to sides  $BC, AC, AB$  at points  $D, E, F$  respectively. Let  $I$  be the incentre of triangle and  $K$  be the intersection point of  $AI, BC$ . Circumcircle  $w$  of the triangle  $\triangle IDK$  intersects  $IB$  and  $IC$  at points  $P$  and  $Q$ . Prove that lines  $EF, PQ$  and the tangent to  $w$  from  $D$  are concurrent.

**§1.7 USAMO 2015, Problem 2**

Quadrilateral  $APBQ$  is inscribed in circle  $\omega$  with  $\angle P = \angle Q = 90^\circ$  and  $AP = AQ < BP$ . Let  $X$  be a variable point on segment  $\overline{PQ}$ . Line  $AX$  meets  $\omega$  again at  $S$  (other than  $A$ ). Point  $T$  lies on arc  $AQB$  of  $\omega$  such that  $\overline{XT}$  is perpendicular to  $\overline{AX}$ . Let  $M$  denote the midpoint of chord  $\overline{ST}$ . As  $X$  varies on segment  $\overline{PQ}$ , show that  $M$  moves along a circle.

**§1.8 All-Russian 2022 Grade 11 Problem 3**

An acute-angled triangle  $ABC$  is fixed on a plane with largest side  $BC$ . Let  $PQ$  be an arbitrary diameter of its circumscribed circle, and the point  $P$  lies on the smaller arc  $AB$ , and the point  $Q$  is on the smaller arc  $AC$ . Points  $X, Y, Z$  are feet of perpendiculars dropped from point  $P$  to the line  $AB$ , from point  $Q$  to the line  $AC$  and from point  $A$  to line  $PQ$ . Prove that the center of the circumscribed circle of triangle  $XYZ$  lies on a fixed circle.

*Terjemahan:* Diberikan sebuah segitiga lancip  $ABC$  yang tetap di sebuah bidang dengan sisi terpanjangnya adalah  $BC$ . Misalkan  $PQ$  adalah sebuah diameter sembarang dari lingkaran luar segitiga  $ABC$ , dengan titik  $P$  berada di busur  $AB$  yang lebih kecil dan titik  $Q$  berada di busur  $AC$  yang lebih kecil. Titik  $X, Y, Z$  berturut-turut adalah kaki tinggi titik  $P$  ke garis  $AB$ , titik  $Q$  ke garis  $AC$ , dan titik  $A$  ke garis  $PQ$ . Buktikan bahwa pusat lingkaran luar segitiga  $XYZ$  berada pada sebuah lingkaran tetap.

**§1.9 All-Russian 2022 Grade 9 Problem 2**

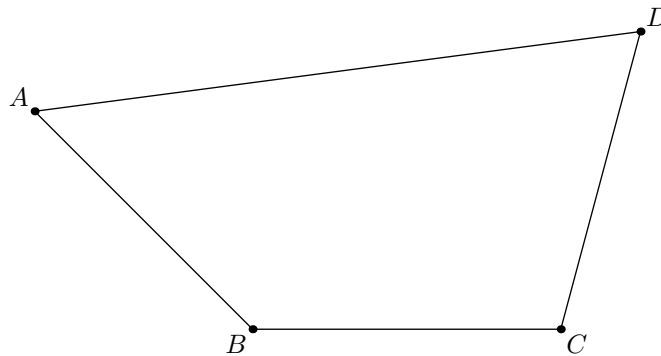
Given triangle  $ABC$  with incenter  $I$  and  $A$ -excenter  $J$ . Circle  $\omega_b$  centered at point  $O_b$  passes through point  $B$  and is tangent to line  $CI$  at point  $I$ . Circle  $\omega_c$  with center  $O_c$  passes through point  $C$  and touches line  $BI$  at point  $I$ . Let  $O_bO_c$  and  $IJ$  intersect at point  $K$ . Find the ratio  $IK/KJ$ .

**§1.10 CGMO 2022, Problem 3**

In triangle  $ABC$ ,  $AB > AC$ ,  $I$  is the incenter,  $AM$  is the midline. The line crosses  $I$  and is perpendicular to  $BC$  intersect  $AM$  at point  $L$ , and the symmetry of  $I$  with respect to point  $A$  is  $J$ . Prove that:  $\angle ABJ = \angle LBI$ .

**§1.11 IMSO 2021**

In the diagram below,  $AB = BC = CD$ ,  $\angle ABC = 135^\circ$  and  $\angle BCD = 105^\circ$ . What is the measure, in degrees, of  $\angle ADC$ ?

**§1.12 Philippine MO 2022, Problem 6**

In  $\triangle ABC$ , let  $D$  be the point on side  $BC$  such that  $AB + BD = DC + CA$ . The line  $AD$  intersects the circumcircle of  $\triangle ABC$  again at point  $X \neq A$ . Prove that one of the common tangents of the circumcircles of  $\triangle BD X$  and  $\triangle CD X$  is parallel to  $BC$ .

**§1.13 Finnish National High School Competition 2019, Problem 3**

Let  $ABCD$  be a cyclic quadrilateral whose side  $AB$  is at the same time the diameter of the circle. The lines  $AC$  and  $BD$  intersect at point  $E$  and the extensions of lines  $AD$  and  $BC$  intersect at point  $F$ . Segment  $EF$  intersects the circle at  $G$  and the extension of segment  $EF$  intersects  $AB$  at  $H$ . Show that if  $G$  is the midpoint of  $FH$ , then  $E$  is the midpoint of  $GH$ .

**§1.14 Finnish National High School Competition 2001, Problem 1**

In the right triangle  $ABC$ ,  $CF$  is the altitude based on the hypotenuse  $AB$ . The circle centered at  $B$  and passing through  $F$  and the circle with centre  $A$  and the same radius intersect at a point of  $CB$ . Determine the ratio  $FB : BC$ .

**§1.15 Iberoamerican 2022, Problem 5**

Let  $ABC$  be an acute triangle with circumcircle  $\Gamma$ . Let  $P$  and  $Q$  be points in the half plane defined by  $BC$  containing  $A$ , such that  $BP$  and  $CQ$  are tangents to  $\Gamma$  and  $PB = BC = CQ$ . Let  $K$  and  $L$  be points on the external bisector of the angle  $\angle CAB$ , such that  $BK = BA, CL = CA$ . Let  $M$  be the intersection point of the lines  $PK$  and  $QL$ . Prove that  $MK = ML$ .

**§1.16 Random Problem by Wildan Bagus Wicaksono**

Diberikan persegi panjang  $ABCD$ . Titik  $P$  adalah perpotongan garis  $BC$  dengan garis yang melalui  $A$  dan tegak lurus  $AC$ . Titik  $Q$  pada segmen  $CD$ . Garis  $PQ$  memotong garis  $AD$  di  $R$ . Garis  $BR$  memotong garis  $AQ$  di  $X$ . Buktikan bahwa jika titik  $Q$  bergerak sepanjang segmen  $CD$ , maka besar  $\angle BXC$  konstan.

(Authored by Wildan Bagus Wicaksono)

**§1.17 USAJMO 2023, Problem 2**

In an acute triangle  $ABC$ , let  $M$  be the midpoint of  $\overline{BC}$ . Let  $P$  be the foot of the perpendicular from  $C$  to  $AM$ . Suppose that the circumcircle of triangle  $ABP$  intersects line  $BC$  at two distinct points  $B$  and  $Q$ . Let  $N$  be the midpoint of  $\overline{AQ}$ . Prove that  $NB = NC$ .

**§1.18 Korea 2023, Problem 1**

In a triangle  $ABC$  ( $\overline{AB} < \overline{AC}$ ), points  $D(\neq A, B)$  and  $E(\neq A, C)$  lies on side  $AB$  and  $AC$  respectively. Point  $P$  satisfies  $\overline{PB} = \overline{PD}, \overline{PC} = \overline{PE}$ .  $X(\neq A, C)$  is on the arc  $AC$  of the circumcircle of triangle  $ABC$  not including  $B$ . Let  $Y(\neq A)$  be the intersection of circumcircle of triangle  $ADE$  and line  $XA$ . Prove that  $\overline{PX} = \overline{PY}$ .

**§1.19 IMO 2023, Problem 2**

Let  $ABC$  be an acute-angled triangle with  $AB < AC$ . Let  $\Omega$  be the circumcircle of  $ABC$ . Let  $S$  be the midpoint of the arc  $CB$  of  $\Omega$  containing  $A$ . The perpendicular from  $A$  to  $BC$  meets  $BS$  at  $D$  and meets  $\Omega$  again at  $E \neq A$ . The line through  $D$  parallel to  $BC$  meets line  $BE$  at  $L$ . Denote the circumcircle of triangle  $BDL$  by  $\omega$ . Let  $\omega$  meet  $\Omega$  again at  $P \neq B$ . Prove that the line tangent to  $\omega$  at  $P$  meets line  $BS$  on the internal angle bisector of  $\angle BAC$ .

**§1.20 Japan 2023, Problem 2**

In an acute triangle  $ABC$ , let  $D, E, F$  be the midpoints of  $BC, CA, AB$  respectively. Let  $X, Y$  be the feet of altitude from  $D$  to  $AB, AC$  respectively. The line through  $F$  parallel to  $XY$  meets line  $DY$  at  $P$ . Prove that  $AD \perp EP$ .

**§1.21 Random Problem 2**

Diberikan segitiga  $ABC$  dengan besar sudut  $\angle ABC = 60^\circ$  dan  $\angle BCA = 100^\circ$ . Pada sisi  $AB$  dan  $AC$  berturut-turut terdapat titik  $D$  dan  $E$  sehingga  $\angle EDC = 2\angle BCD = 2\angle CAB$ . Tentukan besar sudut  $\angle BED$ .

**§1.22 LMNAS SMP 33 Penyisihan nomor 16**

Diberikan  $\triangle ABC$  dengan  $AB = AC$ . Lingkaran dalam  $\triangle ABC$  menyinggung sisi  $AB$  dan  $BC$  berturut-turut di  $P$  dan  $Q$ . Garis  $CP$  memotong lingkaran dalam  $\triangle ABC$  di titik  $P$  dan  $R$ . Jika  $PQ = 4$  dan  $PR = 5$ , panjang  $QR$  adalah ...

- a.  $\frac{1}{2}\sqrt{14}$     b. 2    c.  $\frac{3}{5}\sqrt{14}$     d.  $\frac{4}{5}\sqrt{14}$     e. 3

**§1.23 LMNAS SMP 33 Penyisihan nomor 3 isian singkat**

Diberikan segitiga tumpul  $ABC$  dengan  $AB = BC$ . Titik  $D$  berada di dalam segitiga tersebut sedemikian sehingga  $AD = BD$ ,  $\angle ADB = 140^\circ$ , dan  $\angle ADC = 150^\circ$ . Besar sudut  $ACD$  dalam satuan  $^\circ$  (derajat) adalah ...

**§1.24 LMNAS SMP 31 Semifinal nomor 17**

Diberikan segitiga lancip  $ABC$ . Garis  $\ell_1, \ell_2 \perp AB$  berturut-turut pada  $A$  dan  $B$ . Misalkan  $M$  titik tengah  $AB$ . Garis melalui  $M$  tegak lurus  $AC$  dan  $BC$  memotong  $\ell_1$  dan  $\ell_2$  berturut-turut pada  $E$  dan  $F$ . Misalkan  $D$  titik potong dari  $EF$  dan  $MC$ . Jika  $EF = \frac{64}{3}$ ,  $FM = 16$ , dan  $BD = 12$ . Luas segitiga  $AMF$  dapat dinyatakan dalam  $a\sqrt{b}$  dimana  $a$  dan  $b$  bilangan asli dan  $b$  tidak dapat dibagi oleh bilangan kuadrat sempurna selain 1. Nilai dari  $a + b$  adalah ...

**§1.25 Pelatda Pra OSP SMA 2024 DKI Jakarta**

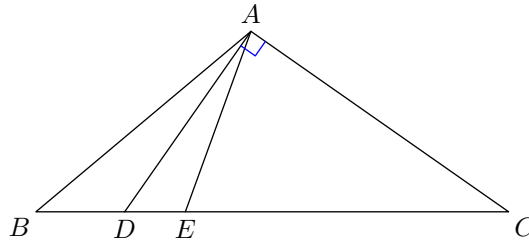
Pada segitiga  $ABC$ , titik  $D$  adalah kaki garis tinggi dari  $C$ . Titik  $E$  dan  $F$  pada  $AC$  dan  $BC$ , berturut-turut  $AE = AD$  dan  $BF = BD$ . Titik  $S$  adalah refleksi  $C$  pada titik pusat lingkaran luar  $\triangle ABC$ . Buktikan bahwa  $SE = SF$ .

**§1.26 IMO 2024, Problem 4**

Let  $ABC$  be a triangle with  $AB < AC < BC$ . Let the incenter and incircle of triangle  $ABC$  be  $I$  and  $\omega$ , respectively. Let  $X$  be the point on line  $BC$  different from  $C$  such that the line through  $X$  parallel to  $AC$  is tangent to  $\omega$ . Similarly, let  $Y$  be the point on line  $BC$  different from  $B$  such that the line through  $Y$  parallel to  $AB$  is tangent to  $\omega$ . Let  $AI$  intersect the circumcircle of triangle  $ABC$  at  $P \neq A$ . Let  $K$  and  $L$  be the midpoints of  $AC$  and  $AB$ , respectively. Prove that  $\angle KIL + \angle YPX = 180^\circ$ .  
(Proposed by Dominik Burek, Poland)

**§1.27 Random Problem 3**

In the diagram below,  $AD$  is perpendicular to  $AC$  and  $\angle BAD = \angle DAE = 15^\circ$ . If  $AB + AE = BC$ , find  $\angle ABC$  in degrees.

**§1.28 KTOM Januari 2025, Nomor 3 Esai**

Diberikan persegi  $ABCD$  dan titik  $E$  pada ruas  $CD$ . Misalkan  $F$  adalah titik pada ruas  $AE$  sehingga  $BF$  tegak lurus  $AE$ . Dengan cara serupa, misalkan  $G$  adalah titik pada ruas  $EB$  sehingga  $AG$  tegak lurus  $EB$ . Misalkan garis  $DF$  dan  $CG$  berpotongan di  $H$ . Buktikan bahwa

$$\angle DFC + \angle DGC = 90^\circ + \angle AHD + \angle BHC.$$

**§1.29 OSN SMP 2016, Nomor 3, Hari Pertama**

Pada segitiga  $ABC$ , titik  $P$  dan  $Q$  berada pada sisi  $BC$  sehingga panjang  $BP$  sama dengan  $CQ$ .  $\angle BAP = \angle CAQ$  dan  $\angle APB$  lancip. Apakah segitiga  $ABC$  sama kaki? Tulis alasan Anda.

**§1.30 Random Problem 4**

Diberikan segitiga  $ABC$  dan lingkaran luar  $\omega$  dari segitiga tersebut. Ditarik sembarang garis yang sejajar  $BC$ , yang memotong sisi  $AB$  dan  $AC$  secara berurutan di titik  $X$  dan  $Y$ . Misalkan  $M$  adalah titik tengah dari busur  $BC$  pada lingkaran  $\omega$  yang tidak memuat titik  $A$ . Garis  $MX$  dan  $MY$

memotong lingkaran  $\omega$  kembali di titik  $S$  dan  $T$  secara berurutan. Misalkan  $ST$  dan  $XY$  berpotongan di titik  $P$ . Tunjukkan bahwa garis  $AP$  menyinggung lingkaran  $\omega$ .

### §1.31 A lemma from Berkeley Math Tournament 2024

The incircle of scalene triangle  $\triangle ABC$  is tangent to  $\overline{BC}$ ,  $\overline{AC}$ , and  $\overline{AB}$  at points  $D$ ,  $E$ , and  $F$ , respectively. The line  $EF$  intersects line  $BC$  at  $P$  and line  $AD$  at  $Q$ . The circumcircle of  $\triangle AEF$  intersects line  $AP$  again at point  $R \neq A$ . If  $I$  is the incenter of  $\triangle ABC$ , prove that  $I, Q, R$  are collinear.

### §1.32 Japan MO 2025, Problem 2

Let  $ABC$  be an acute triangle and let  $O$  be its circumcenter. Let  $O_1$  and  $O_2$  be the circumcenters of triangles  $ABO$  and  $ACO$ , respectively. The circumcircle of triangle  $AO_1O_2$  intersects the side  $BC$  (excluding its endpoints) at two distinct points  $P$  and  $Q$ , such that the four points  $B, P, Q, C$  are collinear in this order. Let  $O_3$  be the circumcenter of triangle  $OPQ$ . Prove that the three points  $A, O, O_3$  are collinear.

### §1.33 OSP SMA 2019, Essay, Problem 5

Diberikan segitiga  $ABC$ , dengan  $AC > BC$ , dan lingkaran luarnya yang berpusat di  $O$ . Misalkan  $M$  adalah titik pada lingkaran luar segitiga  $ABC$  sehingga  $CM$  adalah garis bagi  $\angle ACB$ . Misalkan  $\Gamma$  adalah lingkaran berdiameter  $CM$ . Garis bagi  $\angle BOC$  dan garis bagi  $\angle AOC$  memotong  $\Gamma$  berturut-turut di  $P$  dan  $Q$ . Jika  $K$  adalah titik tengah  $CM$ , buktikan bahwa  $P, Q, O, K$  terletak pada satu lingkaran.

### §1.34 British MO 2019 Round 2, Problem 1

Let  $ABC$  be a triangle. Let  $L$  be the line through  $B$  perpendicular to  $AB$ . The perpendicular from  $A$  to  $BC$  meets  $L$  at the point  $D$ . The perpendicular bisector of  $BC$  meets  $L$  at the point  $P$ . Let  $E$  be the foot of the perpendicular from  $D$  to  $AC$ .

Prove that triangle  $BPE$  is isosceles.

### §1.35 JBMO 2018, Problem 4

Diberikan  $\triangle ABC$  dimana  $A', B', C'$  berturut-turut adalah pencerminan  $A, B, C$  terhadap  $BC, CA, AB$ . Perpotongan lingkaran luar  $\triangle ABB'$  dan  $\triangle ACC'$  adalah  $A_1$ . Definisikan  $B_1$  dan  $C_1$  secara serupa.

Buktikan bahwa  $AA_1$ ,  $BB_1$ , dan  $CC_1$  konkuren (bertemu di satu titik).

### §1.36 Singapore MO Open 2025, Problem 1

In the triangle  $ABC$ ,  $\angle B > 90^\circ$ , the incircle touches the sides  $BC$  and  $CA$  at  $D$  and  $E$ , respectively. The lines  $ED$  and  $AB$  intersect at  $P$ . The incircle of the triangle  $AEP$  touches the sides  $PE$  and  $AP$  at  $D_1$  and  $E_1$ , respectively. The lines  $E_1D_1$  and  $AE$  intersect at  $P_1$ . Suppose  $P, C, E, B$  are concyclic. Prove that  $BE$  is parallel to  $PP_1$ .

### §1.37 Simulasi Semifinal OSN SMP 2025 DKI Jakarta, Uraian, Nomor 1

Diketahui  $ABCD$  adalah segiempat talibusur dimana  $AB$  dan  $CD$  berpotongan di  $E$ . Titik  $K$  dan  $L$  terletak pada  $CD$  dan  $AB$  berturut-turut sehingga  $\angle KAD = \angle KBC$  dan  $\angle LDA = \angle LCB$ . Buktikan bahwa  $EK = EL$ .

### §1.38 Kompetisi Sains Ruangguru 2021 Nomor 1

Diberikan segitiga sama kaki  $ABC$  dengan  $AB = AC$ . Titik  $D$  dan  $E$  berturut-turut merupakan titik tengah  $AC$  dan  $BC$ , dan  $M$  titik tengah  $DE$ . Jika  $AM$  memotong  $BC$  pada  $F$ , buktikan bahwa  $DF$  menyinggung lingkaran luar  $ADM$ .

### §1.39 Iran MO 2018 Round 1, Problem 20

In the convex and cyclic quadrilateral  $ABCD$ , we have  $\angle B = 110^\circ$ . The intersection of  $AD$  and  $BC$  is  $E$  and the intersection of  $AB$  and  $CD$  is  $F$ . If the perpendicular from  $E$  to  $AB$  intersects the perpendicular from  $F$  to  $BC$  on the circumcircle of the quadrilateral at point  $P$ , what is  $\angle PDF$  in degrees?

Pada segiempat siklis  $ABCD$ ,  $\angle B = 110^\circ$ . Perpotongan  $AD$  dan  $BC$  adalah  $E$  dan perpotongan  $AB$  dan  $CD$  adalah  $F$ . Jika garis tegak lurus dari  $E$  ke  $AB$  memotong garis tegak lurus dari  $F$  ke  $BC$  di  $P$ , dengan titik  $P$  terletak pada lingkaran luar segiempat tersebut, berapakah besar  $\angle PDF$  dalam derajat?

### §1.40 Random Problem 5, Classical Incenter Altitude

Diberikan segitiga  $ABC$  dengan  $AB \neq AC$ . Misalkan titik  $I$  adalah pusat lingkaran dalam dan  $r$  adalah panjang jari-jari lingkaran dalam segitiga  $ABC$ . Definisikan  $X$  sebagai titik tengah  $BC$  dan  $Y$  sebagai perpotongan  $XI$  dengan garis tinggi dari  $A$ . Buktikan bahwa panjang  $AY = r$ .

**§1.41 OSS (Seleksi Sekolah) Pradipta Dirgantara 2026**

Diberikan segitiga  $ABC$  dengan  $\angle CAB = 75^\circ$  dan  $ABC = 15^\circ$ . Titik  $D$  pada segmen  $BC$  sehingga  $BD = CA$  dan  $E$  pada sinar  $CA$  sehingga  $CE = BC$ . Misalkan  $F$  titik tengah  $AB$  dan  $M$  perpotongan  $CF$  dengan  $BE$ . Tunjukkan bahwa  $CM$  dan  $DM$  menyinggung lingkaran luar segitiga  $AEF$ .